Course: Intro level astronomy course

Topic: Parallax

Course Level Goal: Students should be able to determine the distance to celestial objects.

Topic Level Goals:

1. Students should be able to explain why measuring distances in space is challenging. They should be able to describe what information they need to measure distance using parallax.

Misconceptions:

Students may not realize how depth perception plays into measuring distance. Students may not understand the limits of parallax and that there are many different ways to measure distance. (the distance ladder, using different methods for different distance ranges)

Assessment:

This can be assessed through i-clicker questions in class. Have students first individually list some thoughts as to why measuring distance in space is challenging (before class). During class after a short demonstration on depth perception the students can get together in groups and try to answer an i-clicker question asking for the distance to some object. Or have the students in groups discuss how they would go about solving such a question.

2. Students should be able to use parallax to measure the distance to a celestial object.

Hard for novice, easy for expert:

To use the parallax equation you will need to measure the angular shift. This sounds like you are measuring an angle but you are measuring how much an object has moved (a line). This might be confusing to students and might lead them to use the wrong units.

Assessment:

This could be a homework problem or a problem on a test where they are asked to measure the distance to Alpha Centauri.

3. Students should be able to examine a hypothetical student's work and determine if it is correct.

Understanding multiple concepts:
A student needs to first read a hypothetical person's work, recognize that is is wrong, and then locate the error and correct it. This requires them to be able to use an understanding of units and

orders of magnitude to check someone’s work and also use an understanding of parallax to correct it.

Assessment:
Students could do this together in class or as test corrections for their own work. For the test corrections they can be required to show where they went wrong and how they know that.